Joint Resource Provisioning and Allocation for Service Function Chaining in Software-Defined Wide Area Networks

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***Abstract*—The software-defined wide area network (SD-WAN) has emerged as a promising architecture for the next-generation WANs, where the Internet service provider (ISP) leases network resources from the infrastructure provider (InP) and offers various applications through service function chaining. Taking the perspective of an ISP, this study focuses on two closely coupled issues. 1) *Resource Allocation (RA):* In bandwidth-constrained SD-WANs, network resources need to be flexibly allocated according to link congestion status. 2) *Resource Provisioning (RP):* Due to service dynamics, ISPs need to adjust their resource leasing schemes in response to request variations. The joint consideration of RA and RP enables bidirectional adaptation between the network and services. In this work, an ILP problem is formulated to unify RA and RP within one framework. For RA, a congestion-aware heuristic algorithm is proposed to achieve load balancing through probabilistic dispersion of data flows across different requests. For RP, a Bayesian Optimization-based algorithm is designed to facilitate resource capacity planning for ISPs. Simulations demonstrate that the proposed methods can reduce request rejection by 50% in bandwidth-constrained networks, improve resource utilization efficiency, enhance the ISP's benefit, and mitigate performance fluctuations caused by service dynamics.**

***Index Terms*—Resource allocation (RA), resource provisioning (RP), service function chaining, software-defined wide area network (SD-WAN).**

I. Introduction

THE ADVENT of software-defined networking (SDN) has heralded a transformative era where network control is decoupled from the underlying hardware. Building upon this paradigm, the concept of software-defined wide area network (SD-WAN) has emerged as a disruptive technology enabling dynamic management of resources across geographically distributed heterogeneous networks, addressing the limitations of traditional wide area networks (WANs), particularly in terms of flexibility and cost-efficiency [1], [2]. As shown in Fig. 1, three entities are involved in a typical SD-WAN paradigm. The infrastructure provider (InP) owns and manages physical network resources. The Internet service provider (ISP) acts as a tenant of the InP, leasing the resource in substrate networks and providing users with a variety of customized applications [3]. Users refer to companies or individuals who submit requests to ISPs and subscribe network services, which often appear in the form of service function chains (SFCs) [4]. For instance, as illustrated by the green arrow in Fig. 1, the video stream generated by a surveillance camera must sequentially traverse the video decoder and pattern recognition services before reaching the console. Taking the perspective of an ISP, this work focuses on the following two issues.

Resource Allocation (RA): The ISP needs to allocate network resources on demand to deliver the requested services, which entails deploying SFCs based on real-time network conditions. Specifically, ISPs must determine optimal locations for deploying each service instance (*Placement*), and assign suitable paths for data flows between adjacent services within the required SFCs (*Routing*).

Resource Provisioning (RP): Given the topology of an SD-WAN, the ISP needs to determine the amount of network resource to lease at each node and link [5], which is referred to as the RP scheme in this article. Furthermore, when significant changes occur in the request distribution, the ISP needs to adjust its RP scheme accordingly, thus adapting to the changing service demands and ensuring the maintenance of service quality.

Unlike the real-time nature of RA, RP operates on a much larger timescale, often spanning days or months. This article primarily focuses on private WAN applications, where the service request distribution evolves in a relatively slow and predictable manner. Consider a corporation with branch offices distributed across multiple regions, changes in its service request patterns are predominantly driven by long-term business developments, such as the establishment of new branches, mergers and acquisitions, seasonal business switches, etc.

A. Challenges in SD-WAN

Both RA and RP face some challenges in SD-WAN scenarios. First, link congestion poses a major challenge to RA. As SD-WAN connects heterogeneous data centers, edge servers, and end-users covering a wide geographical area [6], the bandwidth capacity varies greatly and a majority of the

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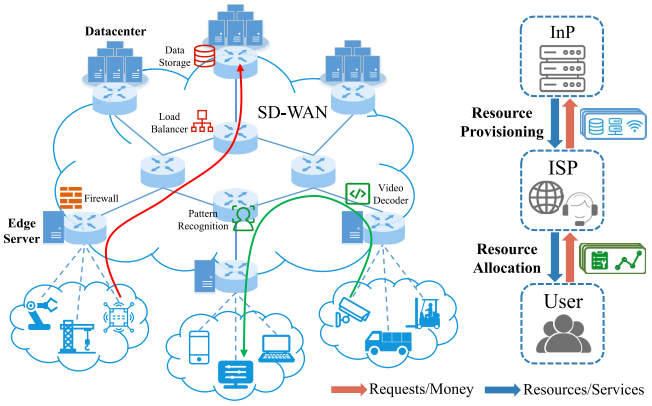

Fig. 1. Illustration of the three roles (InP, ISP, User) and SFCs in SD-WAN.

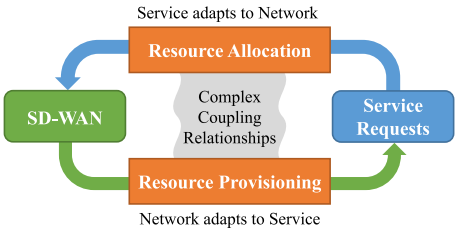

Fig. 2. Bidirectional adaptation between network and service.

links are bandwidth-constrained, especially considering the continuous growth of global IP traffic [7]. Second, as for RP, ISPs usually need to predict the workload distribution on each node and link, thus planning their RP schemes. However, predicting workload distribution based on service request patterns remains challenging. This difficulty stems primarily from the complexity induced by network heterogeneity of SD-WAN [8] and the intricate constraints imposed by service chaining rules and SLAs for diverse personalized applications [9]. These factors collectively render the SFC deployment process highly nonlinear.

In response to the challenges above, we focus on the joint RA and provisioning in SD-WAN, given the inherent interdependence between these two problems. First, since RA involves deploying SFCs under a given RP scheme, the results of RP fundamentally dictate the performance bounds of RA. This relationship is further complicated by the highly nonlinear nature of SFC deployment. As illustrated in Fig. 2, RA and RP complement each other, representing the bidirectional adaptation between network and service. Second, it is essential for ISPs to give equal importance to both RA and RP, as a higher degree of dynamism should be considered in SD-WAN [10], where the services are characterized by high demand for reliability, and the request distribution can undergo significant fluctuations. In such scenarios, neglecting the coupling between RA and RP would inevitably force ISPs into excessive RP to meet service quality guarantees [11]. By jointly addressing RA and RP, we can enhance the efficiency of network resource utilization, thus improving the ISP's revenue.

B. Our Work

In this work, RA and RP are unified within one framework. First, the SFC deployment is formulated as an ILP problem, then we derived a simplified formulation by introducing the routing matrix to streamline *routing*. Based on this formulation, both RA and RP are expressed as parameter-tuning problems. Subsequently, dedicated algorithms are designed following complexity analyses. For RA, a congestion-aware algorithm is designed for the online SFC deployment, where the routing matrix is iteratively updated based on link congestion status. Specifically, data flows from different requests are probabilistically assigned exclusive routing matrices to achieve load balancing, thereby mitigating link congestion and reducing the average end-to-end (E2E) delay. For RP, to help ISPs plan the network capacities, a derivative-free optimization (DFO) method is proposed, where Bayesian optimization (BO) is adopted, converging to the optimal RP scheme by iteratively optimizing the surrogate function. Furthermore, to minimize the impact of RP scheme switching on overall service quality, a smooth switching mechanism is designed to detect significant changes in the request distribution and achieve a smooth transition between RP schemes. Simulation results indicate that our algorithms can improve the efficiency of network resource utilization, increase the acceptance rate of SFC requests, thus enhancing the ISP's revenue. Moreover, the switching of RP schemes effectively mitigates performance fluctuations induced by service dynamics.

The main contributions of this work are listed as follows.

1) An ILP formulation is established considering various refined constraints for SFC deployment in SD-WAN. Subsequently, the routing matrix is introduced to reduce complexity of the original problem.
2) A congestion-aware heuristic algorithm is proposed to flexibly adjust the SFC deployment strategies, which innovatively employs a probabilistic approach to achieve traffic load balancing and reduce link congestion.
3) A BO-based general framework is designed to assist ISPs in planning network resource capacities and adapting to service dynamics. To our knowledge, this is the first work to adopt BO for addressing RP.

The remainder of this article is organized as follows. The related work is reviewed in Section II. Section III presents the ILP formulation and its simplification. In Section IV, RA is solved heuristically, and we solve RP using BO in Section V. Simulation results are analyzed in Section VI, and Section VII summarizes this article.

II. Related Work

A. Service Function Chaining

In the literature, considerable work has been done on the problem of service function chaining [12]. As described in Section I, the deployment of SFCs includes two subproblems: *placement* and *routing*, which are mutually coupled and unable to solve either one without knowing the solution of the other [13]. Basically, the SFC deployment is formulated as an integer or mixed integer linear programming (ILP/MILP) problem [14], [15], [16], where the two subproblems are

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solved together, and many off-the-shelf solvers can be used for its solution, such as Gurobi and CPLEX. However, due to the coupling of *placement* and *routing*, the problem complexity grows exponentially with the network scale [17]. For networks of practical size, solving directly with ILP/MILP solvers usually results in high inefficiency. In light of this, plenty of heuristic approaches are proposed to improve efficiency and feasibility. In [18], *placement* and *routing* are sequentially solved using heuristic methods. By ranking the suitability, the most appropriate node is selected for each service in the requested SFC, and then the Dijkstra algorithm is adopted to select the delay-constrained least-cost (DCLC) path between adjacent services. In [19], greedy algorithms are developed to overcome the MILP's complexity and find efficient routing schemes, minimizing the E2E delay and maximizing the service reliability. In [20], [21], and [22], the authors employed the relaxation-and-rounding strategy to address *placement*. Leveraging the results of *placement*, *routing* can be independently solved through the multicommodity flow models [23].

Note that the above methods typically simplify *routing* by restricting path selection between substrate nodes to single or limited options, potentially overloading high-demand links with congestion risks, particularly in bandwidth-constrained networks such as SD-WAN. Existing studies have focused on mitigating the congestion by performing varied load-balancing mechanisms. In [24], the service placement was modeled as an NP-complete MILP problem, then relaxed into two LP models to address overload/underload on certain service instances. To counter performance degradation from such relaxation in large-scale networks, the authors further proposed a distributed optimization method based on alternating direction method of multipliers (ADMM), featuring rapid convergence and low computational complexity. While this work incorporated link bandwidth constraints, its primary focus remains node-level congestion, i.e., matching service capacity with load demands. In contrast, our study targets bandwidth-constrained networks, emphasizing link-level congestion mitigation. In [25], to address quality degradation caused by overloaded SIP proxies and switches in IP multimedia (MoIP) networks, the SIP message routing problem was formulated as an NP-hard MILP, followed by LP model construction through relaxation. Leveraging the centralized control capability of SDN controllers, this work optimized network traffic holistically to achieve load balancing, thereby enhancing throughput and reducing average latency. However, both studies neglected the refined constraints introduced by SFC, such as SLAs and latency requirements from various personalized applications.

Inspired by [20], [21], and [24], this study simplifies *routing* in SFC deployment through the introduction of routing matrix to reduce computational complexity. Subsequently, to address resultant link congestion caused by this simplification, a heuristic algorithm is proposed for online SFC deployment, where the routing matrix is dynamically updated to achieve traffic load balancing.

B. Resource Provisioning

As for RP, plenty of work has been done in the field of network slicing to address the service dynamics mentioned in Section I [5], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42]. Toosi et al. [26] proposed a dynamic auto-scaling algorithm called ElasticSFC, where flow scheduling and service migration were applied to the deployed SFCs to adapt to changes in service demands. In [27], an optimization framework was proposed for the automatic scaling of network slices based on fluctuating workloads. However, these methods are characterized by adjusting service deployment strategies rather than adjusting network resource capacities, which belongs to the adaptation of services to the network. In [28], a Holt-Winters prediction algorithm was designed to determine traffic demand for each network slice. On this basis, an automatic scaling strategy was proposed, which minimizes resource consumption while ensuring service quality. Similarly, in [29], traffic for network slices is forecasted based on traffic history and user mobility, and provided to a network slice scheduler. However, as stated in Section I, such prediction-based methods are not suitable for the scenario of this article, due to the complexity of SFC deployment in SD-WAN. In [5], [30], and [31], the authors focus on RP for multiple network slices, where various constraints related to SFC deployment are considered, and multiple SFC requests within one slice are aggregated into a Slice Resource Demand group. Note that unlike this work which takes the perspective of an ISP, [5], [30], [31] start from the InP's perspective, planning the resource for each slice with a fixed capacity on each node and link in the substrate network, which can be classified as RA problems.

Recent years have seen a rise in the adoption of machine learning methods to address RP [32], [35], [36]. A systematic literature search leveraging Web of Science with the combined keywords {"machine learning" AND "network RP"} revealed a notable increase in scholarly output, with 480 relevant papers identified between 2017 and 2024, and a remarkable compound annual growth rate (CAGR) of 62.87%. In [33], inspired by advancements in image processing, a deep neural network was designed to predict resource requirements for network slices. In [34], a traffic analysis system was designed utilizing MapReduce to identify traffic patterns within large datasets of traffic traces. Furthermore, many state-of-the-art methods adopt reinforcement learning (RL) or deep RL (DRL) [37], [38], [39], [40], [41], [42], where an agent is trained to choose actions that maximize the InP's profit by analyzing state and reward data obtained through interactions with the environment. However, a significant limitation lies in the demand for extensive computational resources, and the time-consuming nature of model training can hinder the feasibility of these approaches in highly dynamic environments. In this article, BO is adopted for solving RP. Compared to RL and DRL, BO offers a distinct advantage in its simplicity and efficiency due to its full utilization of prior information. Though BO has been widely adopted for hyperparameter tuning in machine learning [43], its applications in network resource

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optimization remain largely confined to wireless RP tasks, such as uplink power control [44], dynamic edge computing management [45], and radio parameter tuning [46]. These efforts primarily focus on the optimization of physical-layer resources (e.g., transmit power, channel allocation) rather than addressing application-layer RP. To the best of our knowledge, this work is the first to bridge BO with application-layer network RP.

In summary, the existing work focusing on RP either adjusts the service deployment instead of adjusting network capacities (e.g., [26] and [27]), or take the perspective of an InP rather than an ISP (e.g., [5], [30], [31], [33], [37], and [38]), or do not consider the various constraints introduced by SFC (e.g., [26], [27], [28], [29], [33], [34], [35], [36], [37], [38], [39], [40], [41], and [42]), or suffer from high computational complexity (e.g., [37], [38], [39], [40], [41], and [42]). To the best of our knowledge, none of the existing works simultaneously overcomes all the aforementioned shortcomings. The goal of this article is to fill this gap by developing a method for solving RP, where 1) Taking the ISP's perspective, we plan network capacities to adapt to service dynamics; 2) Various refined constraints introduced by SFC in SD-WAN are considered; and 3) BO is adopted to overcome the complexity of SFC deployment.

III. Problem Formulation

In this section, we formulate the SFC deployment as an ILP problem. For the simplicity of presentation, the continuous sequence of positive integers $a, a+1, \ldots, b$ is denoted as $[a, b]$.

A. System Model

1) Network Model: The architecture for SFC in SD-WAN is shown in Fig. 3. Fig. 3(a) depicts the example of a substrate network, where we represent the SD-WAN as a unidirectional graph $G(V_s, E_s)$, V_s and E_s are the sets of substrate nodes and links. Considering the heterogeneity of SD-WAN, we assign different resource capacities and prices on each node and link. The capacity of computing resource on node $v \in V_s$ is denoted as c_v, and the price on node v is α_v, meaning that the cost of occupying one unit of computing resource on node v is α_v. Similarly, the capacity and price of bandwidth resource on link $e \in E_s$ are denoted as g_e and β_e. Besides, the delays on each node v and link e are denoted as τ_v and τ_e.

In network G, the set of all possible paths between any two nodes is denoted as $\mathcal{P}$, and the links contained in path $p \in \mathcal{P}$ are denoted by $\Lambda_p = \{e_1, e_2, \ldots\}$. We define mappings from nodes v or links e to paths $p \in \mathcal{P}$: Ψ_v^s (Ψ_v^d) denotes the set of paths starting (ending) with node v, and Ω_e is the set of paths containing link e. Besides, we introduce the term of start-end-node-combination (SENC), I_G represents the set of all SENCs in network G. For each node $v \in V_s$ and SENC $i \in I_G$, $D(i, v) = 1$ if SENC i ends with node v. Further, we define two mappings from node v to SENC i, $S_v = \{i_1, i_2, \ldots\}$ represents the set of SENCs starting with node v, and $D_v = \{i_1, i_2, \ldots\}$ represents the set of SENCs ending with node v.

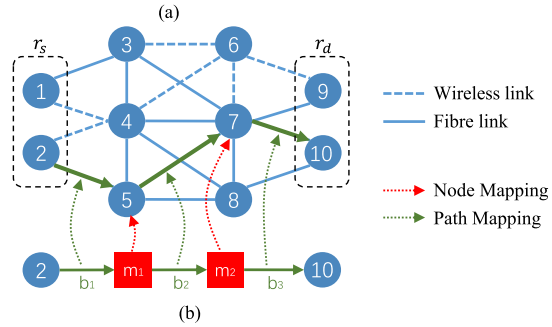

Fig. 3. Example of deploying an SFC request in a substrate network, including the node mapping and path mapping. (a) Substrate network. (b) SFC Request.

2) Service Model: The set of all available services is denoted as M_s, and each type of service $m \in M_s$ causes a certain amount of computing resource consumption when it is deployed on a substrate node. Assume that the computing resource consumed by a service instance is proportional to its workload measured by the input data rate, and service m consumes ϕ_m^{per} computing resource when receiving input data of one unit rate. We use N_v to mark unactivated nodes, additional startup overhead ϕ_v^{up} will be incurred when services are first deployed on these nodes. Fig. 3(b) gives an example of an SFC request $k \in K_r$, which is defined as the following 6-tuple:

$$k = (s_k, d_k, f_k, b_k, \gamma_k, T_k)$$

where $s_k, d_k \in V_s$ are the source and destination nodes. Supposing there are L services in the required SFC, $f_k \in R^{L\times M}$ represents the types of each service, $f_k(l, m) = 1$ if the lth service in the SFC is type m. $b_k \in R^{L\times 1}$ and $b_k(l)$ is the input data rate of the lth service. γ_k denotes the reward obtained by ISP for successfully deploying request k, and the maximum allowable E2E delay is denoted as T_k.

With the notations above, we can derive the amount of computing resource consumed by each service in request k

$$\xi_k(l) = \sum_{m\in M_s} f_k(l, m) \cdot \phi_m^{per} \cdot b_k(l) \quad l = [1, L]. \tag{1}$$

As mentioned in Section I, the source and destination of SFC requests in SD-WAN usually show strong concentration, which is formulated by the Request Distribution (RD). Here, we use a 3-tuple to represent RD of a request batch K_r

$$K_r \sim \text{RD} = (r_s, r_d, \text{rate})$$

where $r_s, r_d \subset V_s$ denote the regions of source and destination nodes, and rate quantifies the regularity with which the source and destination nodes of K_r are distributed. For example, in Fig. 3, $r_s = \{1, 2\}$ and $r_d = \{9, 10\}$, if rate $= 0.75$, then 75% of requests $k \in K_r$ satisfy $s_k \in r_s$ and $d_k \in r_d$ simultaneously.

To describe the SLA constraints, $H_v \in R^{V\times M}$ is used to indicate the service types allowed to be deployed on each node, $H_v(v, m) = 1$ if node v is allowed to deploy service instances of type m. Similarly, $H_p \in R^{P\times M}$ and $H_p(p, m) = 1$ if path p is capable of delivering the output data flow of service m.

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TABLE I
DEFINITIONS OF NOTATIONS

Network Related Notations	
$G(V_s, E_s)$	Topology of the substrate network
$\mathcal{P}$	Set of all possible paths between any two nodes in G
Λ_p	Set of links contained in path p
Ψ_v^s, Ψ_v^d	Set of paths starting/ending with node v
Ω_e	Set of paths containing link e
I_G	Set of start-end-node-combination(SENC) in G
D	$D(i, v) = 1$ if SENC i end with node v
S_v, D_v	Set of SENCs starting/ending with node v
c_v, α_v	The amount and price of computing resource on node v
g_e, β_e	The amount and price of bandwidth resource on link e
τ_v, τ_e	Delay experienced at node v and link e
N_v	$N_v = 1$ if no service has been deployed on node v
ϕ_v^{up}	Additional startup overhead of node v
H_v	$H_v(v, m) = 1$ if node v is capable of deploying services of type m
H_p	$H_p(p, m) = 1$ if path p is capable of delivering the output data flow of service m
Service Related Notations	
M_s	Set of all service types
ϕ_m^{per}	Computing resource consumed by service of type m to serve one unit load
K_r	Set of all SFC requests
f_k	$f_k(l, m) = 1$ if the lth service in request k is type m
s_k, d_k	Source and destination nodes of request k
b_k	$b_k(l)$ is the input data rate of the lth service in request k
ξ_k	$\xi_k(l)$ is the computing resource consumed by the lth service in request k
γ_k	Reward of deploying request k
T_k	Maximum allowable E2E delay of request k
r_s, r_d	Region of source and destination nodes of K_r

B. Optimization Formulation

The notations used to formulate the SFC deployment in SD-WAN are defined in Table I.

Variables: Fig. 3 shows the result of deploying a requested SFC onto the substrate network. Each service in the SFC can be regarded as a virtual node, and the data flows can be regarded as virtual links. The two subproblems *placement* and *routing* can be regarded as mapping each virtual node and link in the SFC request to a substrate node and path, respectively. To solve the two subproblems together, the following variables are introduced.

1) $z_v(l, v, k)$: Corresponding to the *node mapping*. Boolean variable that equals 1 if the lth node in the SFC is mapped to node $v \in V_s$, $l = [0, L+1]$.
2) $z_p(l, p, k)$: Corresponding to the *path mapping*. Boolean variable that equals 1 if the lth link in the SFC is mapped to path $p \in \mathcal{P}$, $l = [1, L+1]$.
3) a_k: Boolean variable that equals 1 if request k is accepted.
4) x_v: Boolean variable that equals 1 if node v is occupied to deploy the current batch of requests.

Objective: The optimization objective is to maximize the ISP's benefit, which is the difference between Revenue R and Cost C, where R is the sum of the reward of each accepted request, and C includes the cost of computing and bandwidth resources, which can be derived from c_v^{use} and g_e^{use}, the resources consumed on each node and each link

$$\max_{a,x,z_v,z_p} \quad R - C \tag{2}$$

$$R = \sum_{k \in K_r} a_k \cdot \gamma_k \tag{3}$$

$$C = \sum_{v \in V_s} \alpha_v \cdot c_v^{\text{use}} + \sum_{e \in E_s} \beta_e \cdot g_e^{\text{use}} + \sum_{v \in V_s} x_v \cdot N_v \cdot \phi_v^{up} \tag{4}$$

$$c_v^{\text{use}} = \sum_{k \in K_r} \sum_{l=1}^{L} z_v(l, v, k) \cdot \xi_k(l) \quad v \in V_s \tag{5}$$

$$g_e^{\text{use}} = \sum_{k \in K_r} \sum_{p \in \Omega_e} \sum_{l=1}^{L+1} z_p(l, p, k) \cdot b_k(l) \quad e \in E_s. \tag{6}$$

Constraints:

$$c_v^{\text{use}} \le c_v \qquad \forall v \in V_s \tag{7}$$

$$g_e^{\text{use}} \le g_e \qquad \forall e \in E_s \tag{8}$$

$$z_v(0, s_k, k) = a_k, \ z_v(L+1, d_k, k) = a_k \quad \forall k \in K_r \tag{9}$$

$$\sum_{p \in \Psi_{s_k}^s} z_p(1, p, k) = a_k, \ \sum_{p \in \Psi_{d_k}^d} z_p(L+1, p, k) = a_k \quad \forall\ k \in K_r \tag{10}$$

$$\sum_{p \in \Psi_v^d} z_p(l, p, k) - \sum_{p \in \Psi_v^s} z_p(l+1, p, k) = 0 \quad \forall v \in V_s \ \ \forall k \in K_r \ \ \forall l = [1, L] \tag{11}$$

$$\sum_{l=1}^{L} \sum_{v \in V_s} z_v(l, v, k) \cdot \tau_v + \sum_{l=1}^{L+1} \sum_{p \in \mathcal{P}} \sum_{e \in \Lambda_p} z_p(l, p, k) \cdot \tau_e \le T_k \quad \forall k \in K_r \tag{12}$$

$$\sum_{m \in M_s} \sum_{v \in V_s} z_v(l, v, k) \cdot H_v(v, m) \cdot f_k(l, m) = a_k \quad \forall k \in K_r \ \ \forall l = [1, L] \tag{13}$$

$$\sum_{m \in M_s} \sum_{p \in \mathcal{P}} z_p(l, p, k) \cdot H_p(p, m) \cdot f_k(l, m) = a_k \quad \forall k \in K_r \ \ \forall l = [2, L+1] \tag{14}$$

$$\sum_{v \in V_s} z_v(l, v, k) = a_k \quad \forall k \in K_r \ \ \forall l = [0, L+1] \tag{15}$$

$$\sum_{p \in \mathcal{P}} z_p(l, p, k) = a_k \quad \forall k \in K_r \ \ \forall l = [1, L+1] \tag{16}$$

$$\theta_v = \frac{c_v^{\text{use}}}{c_v} \le x_v \quad \forall v \in V_s. \tag{17}$$

Equations (7) and (8) ensure that the resource usage on each node and link would not exceed its capacity. Equations (9) and (10) ensure that the data flow starts from s_k and ends at d_k. Equation (11) means that for any two adjacent links in the required SFC, the end of the previous link and the start of the latter one must be mapped to the same substrate node, which ensures the continuity between data flows. Equation (12) ensures that the E2E delay of request k, including the delay experienced on each node and link, must not exceed T_k. Equations (13) and (14) ensure that all the required services must be deployed on capable substrate nodes, and select a path capable of delivering its output data flow. Equations (15) and (16) ensure that for each service (or virtual link) in the required SFC, up to one substrate node (or path) can be

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selected for its deployment. Equation (17) means that for all occupied nodes v, x_v is set to 1.

Summarizing the aforementioned definitions and constraints, we can formulate the ILP problem as below, denoted as *PR* (Placement and Routing)

$$\begin{aligned} \max_{a,x,z_v,z_p} \quad & R - C \\ \text{s.t.} \quad & \text{constraints} \quad (7)-(17) \\ \text{vars:} \quad & a_k \in \{0,1\} \quad \forall k \in K_r \\ & x_v \in \{0,1\} \quad \forall v \in V_s \\ & z_v(l,v,k) \in \{0,1\} \quad \forall k \in K_r, v \in V_s, l = [0, L+1] \\ & z_p(l,p,k) \in \{0,1\} \quad \forall k \in K_r, p \in \mathcal{P}, l = [1, L+1]. \end{aligned} \tag{18}$$

In PR (18), *placement* and *routing* are solved together, this formulation can be reduced to the classical graph embedding problem, which has been proven NP-hard and inapproximable [47]. Another challenge lies in the extensive scale, as all possible paths between each SENC are considered [13], forming the path set $\mathcal{P}$. As the network scale grows, the size of $\mathcal{P}$ increases exponentially, making it hard for PR to be applied to practical-size problems. However, the vast majority of $\mathcal{P}$ is impractical for carrying the data flow between adjacent services in an SFC. Therefore, in the following sections, PR is simplified by considering only one path for each SENC.

C. Problem Simplification

Aiming to reduce the complexity, we simplify PR (18) by selecting only one path for each SENC. These selected paths are summarized in the *routing matrix* $P \in R^{I \times E}$, $P(i,e) = 1$ if link e is on the path assigned for SENC i. The introduction of routing matrix establishes a connection between node v and path p, thereby enabling the unification of the node and path mapping in PR as a single mapping, namely the SENC mapping. Therefore, $z_v(l,v,k)$ and $z_p(l,p,k)$ in PR (18) can be replaced by one boolean variable $y(l,i,k)$, corresponding to the mapping between each data flow in the requested SFC to a SENC $i \in I_G$ in the substrate network. $y(l,i,k) = 1$ if the lth virtual link in request k is mapped to SENC i ($l = [1, L+1]$).

For example, as depicted in Fig. 4, let i_{26} denote the SENC of node 2 and 6, given an SFC request k including two services, $y(2, i_{26}, k) = 1$ means that the second data flow in request k is mapped to SENC i_{26}. On one hand, service m_1 and m_2 should deploy on nodes 2 and 6, respectively. On the other hand, the data flow between m_1 and m_2 should go through $Path(i_{26}) = \{e | P(i_{26}, e) = 1\}$. In Fig. 4, $Path(i_{26}) = \{e_{25}, e_{56}\}$, so the data flow goes through nodes $2 \to 5 \to 6$ sequentially.

Based on the optimization variable y, two expressions can be derived

$$y_e(l,e,k) = \sum_{i \in I_G} y(l,i,k) \cdot P(i,e). \tag{19}$$

$y_e(l,e,k) = 1$ if link e is chosen to transfer the data flow in the lth link of SFC k

$$y_v(l,v,k) = \sum_{i \in I_G} y(l,i,k) \cdot D(i,v). \tag{20}$$

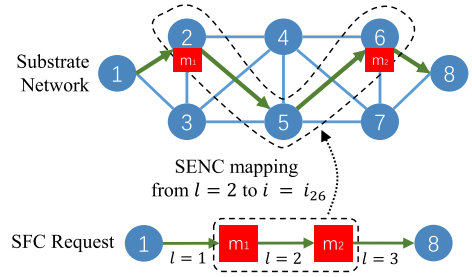

Fig. 4. Illustrative example of SENC Mapping.

$y_v(l,v,k) = 1$ if node v is chosen to deploy the lth service in SFC k. Another modified notation is H_p in (14), which is replaced by $H_i \in R^{I \times M}$, and $H_i(i,m) = 1$ if SENC i is capable of delivering the output data flow of service m. With the modifications above, PR can be simplified as follows.

Objective:

$$\max_{a,x,y} \quad R - C \tag{21}$$

$$R = \sum_{k \in K_r} a_k \cdot \gamma_k \tag{22}$$

$$C = \sum_{e \in E_s} \beta_e \cdot g_e^{\text{use}} + \sum_{v \in V_s} \alpha_v \cdot c_v^{\text{use}} + \sum_{v} x_v \cdot N_v \cdot \phi_v^{up} \tag{23}$$

$$c_v^{\text{use}} = \sum_{k \in K_r} \sum_{l=1}^{L} y_v(l,v,k) \cdot \xi_k(l) \quad v \in V_s \tag{24}$$

$$g_e^{\text{use}} = \sum_{k \in K_r} \sum_{l=1}^{L+1} y_e(l,e,k) \cdot b_k(l) \quad e \in E_s. \tag{25}$$

Constraints:

$$c_v^{\text{use}} \le c_v \quad \forall v \in V_s \tag{26}$$

$$g_e^{\text{use}} \le g_e \quad \forall e \in E_s \tag{27}$$

$$\sum_{i \in S_{s_k}} y(1,i,k) = a_k \quad \forall k \in K_r \tag{28}$$

$$\sum_{i \in D_{d_k}} y(L+1,i,k) = a_k \quad \forall k \in K_r \tag{29}$$

$$\sum_{i \in D_v} y(l,i,k) - \sum_{i \in S_v} y(l+1,i,k) = 0 \quad \forall v \in V_s \quad \forall k \in K_r \quad \forall l = [1, L] \tag{30}$$

$$\sum_{l=1}^{L} \sum_{v \in V_s} y_v(l,v,k) \cdot \tau_v + \sum_{l=1}^{L+1} \sum_{e \in E_s} y_e(l,e,k) \cdot \tau_e \le T_k \quad \forall k \in K_r \tag{31}$$

$$\sum_{m \in M_s} \sum_{v \in V_s} y_v(l,v,k) \cdot H_v(v,m) \cdot f_k(l,m) = a_k \quad \forall k \in K_r \quad \forall l = [1, L] \tag{32}$$

$$\sum_{m \in M_s} \sum_{i \in I_G} y(l,i,k) \cdot H_i(i,m) \cdot f_k(l,m) = a_k \quad \forall k \in K_r \quad \forall l = [2, L+1] \tag{33}$$

$$\sum_{i \in I_G} y(l,i,k) = a_k \quad \forall k \in K_r \quad \forall l = [1, L+1] \tag{34}$$

$$\theta_v = \frac{c_v^{\text{use}}}{c_v} \le x_v \quad \forall v \in V_s. \tag{35}$$

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The simplified PR (SPR) problem is summarized as follows:

$$\begin{aligned} \max_{a,x,y} \quad & R - C \\ \text{s.t.} \quad & \text{constraints} \quad (26)-(35) \\ \text{vars:} \quad & a_k \in \{0,1\} \quad \forall k \in K_r \\ & x_v \in \{0,1\} \quad \forall v \in V_s \\ & y(l,i,k) \in \{0,1\} \forall k \in K_r \quad \forall i \in I_G \quad \forall l = [1, L+1]. \end{aligned} \tag{36}$$

The complexity of SPR (36) is mainly determined by the scales of y and P, which grow polynomially with the network size, since $|I_G| \leq V^2$, where V and $|I_G|$ are the amount of nodes and SENCs in G. Besides, as $y \in R^{L \times |I_G| \times K}$, the complexity of SPR can be denoted as $O(K \cdot L \cdot V^2)$. Compared to PR (18), SPR is better suited for solving SFC problems in practical-size networks.

D. Two Parameter Tuning Problems

In a practical scenario, the SFC requests arrive sequentially, when the number of pending requests reaches the batch size K, the ISP solves SPR (36) and deploys the current batch of requests based on optimization results. In SPR, there are two types of adjustable parameters, namely the routing matrix P and network capacities (c_v, g_e). When the above parameters are fixed, the process of solving SPR is denoted as follows:

$$\left(a^*, x^*, y^*\right) = \text{SPR}(P, c_v, g_e). \tag{37}$$

To ensure the quality of service (QoS) and maximize benefit, the ISP should carefully tune the P and (c_v, g_e) during the deployment process. These two parameter-tuning problems correspond to RA and RP in Section I, respectively.

1) *RA:* The ISP needs to adjust the routing matrix P based on the current congestion state of network G. This enables the SFC deployment to adapt to the dynamically changing network.
2) *RP:* When significant changes occur in RD, the ISP needs to adjust its RP scheme, that is, the network capacities (c_v, g_e). This enables the network to dynamically adapt to evolving service demands.

In summary, the bidirectional adaptation between network and service requires tuning two types of parameters in SPR.

IV. Resource Allocation

In this section, the formulation of RA is derived. Subsequently, an analysis is conducted to assess its complexity, followed by the design of a heuristic algorithm.

A. Resource Allocation Problem

Since only one path is specified for each SENC in SPR (36), some popular links may appear in the paths of multiple SENCs. As the SFC deployment proceeds, these popular links are prone to get congested, preventing the network from accepting subsequent requests and deteriorating the ISP's benefit. This issue becomes particularly critical when the network is bandwidth-constrained and SFC requests are intensively distributed, i.e., rate in RD is close to 1. Therefore, the ISP needs to dynamically adjust the routing matrix P during the deployment process. By improving the flexibility of chaining rules for each SFC request, the ISP can achieve load balancing for data flows and alleviate network congestion.

When tuning the routing matrix, the ISP aims to minimize *Congest*, the overall link congestion across the whole network. Meanwhile, the link delay should also be taken into account. Therefore, the optimization object of RA is denoted as follows:

$$Obj = w_1 \cdot Congest + w_2 \cdot LinkDelay \tag{38}$$

where w_1 and w_2 are adjustable weight parameters. First, given the routing matrix P, *LinkDelay* is the sum of the E2E delay for each SENC $i \in I_G$, representing the overall link delay across the substrate network

$$LinkDelay = \sum_{i \in I_G} \sum_{e:P(i,e)=1} \tau_e. \tag{39}$$

Second, the overall congestion *Congest* is derived as follows:

$$Congest = \sum_{e \in E_s} \text{Con}\left(\theta_e^{\text{new}}\right) \tag{40}$$

where $\text{Con}(\theta_e^{\text{new}})$ denotes the congestion degree on each substrate link $e \in E_s$, characterized by θ_e^{new}, the bandwidth utilization on link e after deploying the current batch of requests. Specifically, we adopt the following piece-wise function:

$$\text{Con}\left(\theta_e^{\text{new}}\right) = \begin{cases} 1, & \text{if} \quad 0 \leq \theta_e^{\text{new}} \leq \theta_1 \\ 1+\epsilon, & \text{if} \quad \theta_1 < \theta_e^{\text{new}} \leq \theta_2 \\ 2+\epsilon, & \text{if} \quad \theta_2 < \theta_e^{\text{new}} \leq 1 \end{cases} \tag{41}$$

where ϵ is an adjustable positive constant, $0 < \theta_1 < \theta_2 < 1$ are two adjustable thresholds.

Here, we analyze the impact of the routing matrix P on the overall network congestion degree *Congest*. The relationship between P and *Congest* can be summarized as follows:

$$P \rightarrow y^* \rightarrow g_e^{\text{use}} \rightarrow \theta_e^{\text{new}} \rightarrow Congest.$$

Assuming that P is fixed, the optimal solution y^* can be obtained by solving SPR (36)

$$\begin{aligned} & y^*(l,i,k) = \text{SPR}(P) \\ & \forall k \in K_r \quad \forall i \in I_G \quad \forall l = [1, L+1]. \end{aligned} \tag{42}$$

Based on y^*, we can derive g_e^{use}, the amount of bandwidth occupied by deploying the current batch of requests

$$g_e^{\text{use}} = \sum_{k \in K_r} \sum_{i \in I_G} \sum_{l=1}^{L+1} y^*(l,i,k) \cdot D(i,v) \quad \cdot \xi_k(l) \qquad \forall e \in E_s. \tag{43}$$

Then, we can derive θ_e^{new} as follows:

$$\theta_e^{\text{new}} = \theta_e^{\text{old}} + \frac{g_e^{\text{use}}}{g_e} \qquad \forall e \in E_s \tag{44}$$

where θ_e^{old} is the bandwidth utilization on link e before deploying the current batch of requests. Finally, the overall network congestion degree *Congest* can be derived from θ_e^{new}, as shown in (40). Besides, there are some constraints on P itself. Assume that the start and end nodes of SENC i are v_1, v_2, and there are N_i feasible paths from v_1 to v_2, then these

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paths can be denoted by $\text{Path}_i \in R^{N_i \times E}$, $\text{Path}_i(n, e) = 1$ if link e is on the nth path of SENC i. Since the path specified for each SENC i must be feasible, the ith row of P, denoted as P_i, must be a row vector selected from $Path_i$. So we have

$$P_i \in \text{Row}(\text{Path}_i) \qquad \forall i \in I_G. \tag{45}$$

With the above analysis, RA is summarized as follows:

$$\begin{aligned} \min_{P} \quad & \text{Obj} = w_1 \cdot Congest + w_2 \cdot LinkDelay \\ \text{s.t.} \quad & \text{constraints} \quad (39)-(45) \\ \text{vars:} \quad & P(i, e) \in \{0, 1\} \qquad \forall i \in I_G \quad \forall e \in E_s. \end{aligned} \tag{46}$$

The above problem is difficult to solve mainly due to the following reasons.

Severe Nonlinearity: First, as a piecewise function, Con in (40) introduces nonlinearity to RA. Second, in constraint (42), the process of solving the SPR problem is abstracted as function SPR with P as input, and there is no analytical expression for the relationship between P and y^*. So the RA formulation is characterized by severe nonlinearity, making it unsuitable for existing ILP/MILP solvers.

Extensive Scale: Generally, DFO methods are applicable for solving problems that lack explicit expressions between the optimization variables and objects. However, DFO methods are not suitable for solving RA due to their extensive scale. The dimension of P grows geometrically with the network size, rendering the solution space excessively large for problems of realistic scale. Additionally, considering the online deployment of SFC requests, the resolution of RA necessitates a high level of real-time performance. Therefore, DFO methods are difficult to converge within an acceptable time for RA.

Consequently, in the following section, we propose a heuristic algorithm to solve RA.

B. Heuristic Approach

To adapt the SFC deployment strategies to changing networks, the ISP needs to dynamically adjust the routing matrix P during the deployment process. On the one hand, each time before deploying an incoming batch of SFC requests, the ISP needs to adjust P according to the current congestion status of the substrate network. On the other hand, the congestion on certain popular links may intensify when a batch of SFC requests share the same routing matrix. This issue becomes particularly prominent with the increase of rate and batch size K. Therefore, it is necessary to diffuse the data flow from different requests within the same batch.

From the above two considerations, a heuristic algorithm is designed as shown in Algorithm 1. The algorithm is called iterative batch deployment (IBD), because P is iteratively updated during the deployment of each batch of SFC requests. Specifically, the iterations of P are divided into the following two categories.

1) Outer Iteration: In the RA formulation (46), P is tuned to reduce the overall network congestion *Congest*, which is the sum of the congestion degrees on each link, i.e., $\text{Con}(\theta_e)$. Therefore, in Algorithm 1, we take $\text{Con}(\theta_e)$ as the weight w_e of each substrate link $e \in E_s$. In the initial state, $\theta_e = 0$ and

Algorithm 1: Iterative Batch Deployment Algorithm (IBD)

1 initialize network parameters: c_v, g_e, N_v, P_{ini};
2 set $w_e = 1$ for all $e \in E_s$, $prob(i) = 0.5$ for all $i \in I_G$;
3 **for** *each batch of requests in* $\{K_r\}$ **do**
4 load the current batch of requests: $\{K_r^{now}\}$;
5 calculate link weights: $w_e = Con(\theta_e)$;
6 update the routing matrix: $P_{now} = ShortPath(G, w_e)$;
7 set $InnerIterStop = 0$, $P_{new} = P_{now}$;
8 **while** $InnerIterStop = 0$ **do**
9 $Obj_0 \leftarrow w_1 \cdot Congest + w_2 \cdot LD$;
10 calculate exclusive routing matrix P_k for each request $k \in K_r^{now}$ according to *prob*;
11 solve the SPR problem: $(a^*, x^*, y^*) = SPR(P_k)$;
12 evaluate the results of this iteration: $(w_e, P_{new}, prob, Obj) = Evaluate(P_{now}, G, y^*, g_e^{now}, g_e^{ini})$;
13 **if** $(Obj \le Obj_0) OR (n_{iter} \ge n_{\max})$ **then**
14 $InnerIterStop \leftarrow 1$;
15 **end**
16 **end**
17 deploy the current batch of requests according to optimization results (a^*, x^*, y^*);
18 update network status parameters: c_v, g_e, N_v;
19 **end**

$w_e = 1$ for each link e, and the routing matrix P_{ini} is initialized by computing the shortest path for each $i \in I_G$.

As shown in lines 5 and 6 in Algorithm 1, before deploying each batch of requests, the ISP updates the weight of each substrate link according to its current congestion degree and then updates the routing matrix by calculating the weighted shortest path for each $i \in I_G$

$$P_{\text{now}} = ShortPath(G, w_e). \tag{47}$$

In this way, P can be adjusted according to the link congestion before deploying each batch of requests, enabling data flows to "bypass" those already congested links.

2) Inner Iteration: The inner iteration corresponds to lines 7–16 of Algorithm 1. We hope to disperse the data flow from different requests, thus utilizing the bandwidth on each link as evenly as possible. In other words, for the same SENC i, different requests $k \in K_r^{\text{now}}$ can choose different paths. Therefore, an exclusive routing matrix P_k is assigned to request k. Accordingly, when solving SPR (36), (19) is modified as follows:

$$y_e(l, e, k) = \sum_{i \in I_G} y(l, i, k) \cdot P_k(i, e). \tag{48}$$

As shown in line 9 of Algorithm 1, the IBD algorithm aims to reduce *Obj* within the inner iteration. Similar to (38) in RA, both the congestion degree *Congest* and link delay *LD* are considered in *Obj*, with w_1 and w_2 as their weights. As shown in lines 11 and 12, the SPR problem is solved using the current set of routing matrices P_k, then we evaluate the optimization results y^*, update w_e and *Obj*.

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Function $Evaluate(P_{now}, G, y^*, g_e^{now}, g_e^{ini})$
  calculate g_e^{use} using y^* according to (43)
  calculate link utilization rate for all $e \in E_s$:
    $\theta_e = \left(g_e^{ini} - g_e^{now} + g_e^{use}\right)/g_e^{ini}$
  update link weights: $w_e = Con(\theta_e)$
  calculate routing matrix $P_{new} = ShortPath(G, w_e)$
  for *each* $i \in I_G$ **do**
    get bb_i^{old}, the bottleneck bandwidth according to the original routing path in P_{now}
    get bb_i^{new}, the bottleneck bandwidth according to the new routing path in P_{new}
    $prob(i) = \frac{bb_i^{new}}{bb_i^{old} + bb_i^{new}}$
  calculate *Congest* and *LD* using the latest w_e and *prob*
  update *Obj*: $Obj = w_1 \cdot Congest + w_2 \cdot LD$
Output: $w_e, P_{new}, prob, Obj$

Function *Evaluate* shows the evaluation of the optimization results. First, assume that the current batch of requests has been deployed according to y^*, then the link weights w_e are updated and a new routing matrix P_{new} is calculated based on the latest w_e. P_{now} and P_{new} represent the path selection based on the congestion state before and after the current iteration, respectively. We update the routing matrix P_k for each $k \in K_r^{\text{now}}$ by combining P_{now} and P_{new} in a probabilistic way. For each SENC $i \in I_G$, request k selects the new path with a certain probability prob(i). Therefore, for each request $k \in K_r^{\text{now}}$, the ith row of P_k picks the ith row in P_{new} and P_{now} with probabilities prob(i) and $1-\text{prob}(i)$, respectively. prob(i) should be carefully tuned to balance the load allocated to the old and new paths. In this article, the probability of choosing the old and new paths is proportional to their bottleneck bandwidths, denoted as bb_i^{old} and bb_i^{new}

$$\frac{\text{prob}(i)}{1-\text{prob}(i)} = \frac{bb_i^{\text{new}}}{bb_i^{\text{old}}} \quad \begin{cases} bb_i^{\text{new}} = \min_{e:P_{\text{new}}(i,e)=1} b_e \\ bb_i^{\text{old}} = \min_{e:P_{\text{now}}(i,e)=1} b_e. \end{cases}$$

In this way, load balancing for data flows from different requests is achieved in a probabilistic manner. Accordingly, with the probability prob, the link delay in *Obj* is adjusted as follows:

$$\text{LD} = \sum_{i \in I_G} \left[\text{prob}(i) \cdot \sum_{e:P_{\text{new}}(i,e)=1} \tau_e + (1-\text{prob}(i)) \cdot \sum_{e:P_{\text{now}}(i,e)=1} \tau_e \right]. \quad (49)$$

As shown in lines 13–15 of Algorithm 1, after the evaluation of y^*, the inner iteration is continued or stopped depending on whether y^* would yield an increase in Obj. If there is an increase of Obj, load balancing is required for K_r^{now}, and the inner iteration continues. Otherwise, the iteration ends and the current optimization results are adopted to deploy K_r^{now}. Besides, we set $n_{\max}$ as the maximum number of inner iterations to ensure the convergence of IBD.

C. Complexity and Performance Analysis

Complexity: Compared to the original ILP formulation (PR), IBD reduces the computational complexity by solving SPR, while alleviating link congestion through updating the routing matrix in inner and outer iterations. Due to the complexity of SPR being $O(K \cdot L \cdot V^2)$, the complexity of IBD is affected by the network scale, with its execution time proportional to V^2, which limits the scalability of IBD in large-scale networks. This is because paths between each SENC $i \in I_G$ need to be calculated to form the routing matrix, resulting in the scale of P and y in SPR being proportional to V^2.

However, note that for concentrated request distribution patterns, the path selection exhibits strong sparsity characteristics, meaning the majority of the V^2 paths in the routing matrix will not appear in the solution. Therefore, a potential complexity reduction approach is on-demand routing, where K E2E paths are calculated for the K requests, forming a smaller routing matrix $SP \in R^{K \times E}$. This makes the deployment algorithm's complexity independent of network scale V, significantly enhancing its scalability in large-scale networks. We refer to the above deployment algorithm as Fast IBD (FIBD). Compared to IBD, FIBD fundamentally trades off solution accuracy for computational efficiency via further routing simplification. Therefore, maintaining performance parity with IBD requires designing a more sophisticated load-balancing mechanism for FIBD. For instance, K_r^{now} can be grouped by source-destination pairs. Within each group, inter-request load balancing follows the probabilistic approach in IBD: different E2E paths are probabilistically assigned to individual requests while maintaining group-level path diversity.

Performance Bound: To establish the performance bound of IBD, we analyze its worst-case scenario where IBD fails to implement load balancing. Specifically, under extreme network conditions (e.g., star-shaped topology), the routing matrices P_{now} and P_{new} in line 12 of Algorithm 1 become identical, indicating no alternative routing options. In this degenerate scenario, the inner iterations of IBD become invalid, and all requests within one batch share identical routing matrices. To validate the worst-case scenario through simulation, the Simplified IBD (SIBD) algorithm is designed by removing the inner iterations, i.e., $n_{\max}$ is set to 1. In Section VI, SIBD serves as a baseline for comparison with IBD, thereby empirically quantifying IBD's performance boundaries.

D. Practical Implementation

Fig. 5 illustrates the implementation of RA (46) in a practical SD-WAN deployment scenario. First, based on the user's functional requirements and available service instance information, the service orchestrator generates structured SFC requests (6-tuple $k \in K_r$), which serve as the input of RA. Then, the ISP solves RA considering network resource and SLA constraints. As previously discussed, the solution $y(l, i, k)$ can be divided into two parts, representing node mapping (y_v) and link mapping (y_e). y_v is provided to container orchestration systems (e.g., Kubernetes) to configure and manage containerized services on substrate nodes; while y_e instructs SDN controllers (e.g., ONOS, Ryu, OpenDaylight) to program flow tables and steer SFC traffic through designated paths.

To configure routing policies on each SDN-enabled node, y_v and y_e derived from the solutions of RA need to be converted

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into standardized flow entries compliant with SDN control protocols. The essential configuration parameters at each node comprise key match fields: source/destination IP addresses and input/output port numbers, while other parameters (e.g., AppID, Priority) fall beyond our research scope. The conversion from RA solutions to flow table parameters requires two mappings: IP-node mapping $\Gamma(v)$ and port-link mapping $\Pi(e)$. Specifically, as depicted in Fig. 6, for each accepted SFC request, the substrate path for each virtual link needs to be mapped, which requires: (i) determining the substrate node set $V^* = \{v_1, v_2, \ldots, v_J\}$, and (ii) identifying the substrate link set $E^* = \{e_1, e_2, \ldots, e_{J+1}\}$. Then, on each substrate node $v^* \in V^*$, srcIP and dstIP are assigned as $\Gamma(v^*)$, while InPort and OutPort are assigned as $\Pi(e^*)$.

In hybrid network scenarios, ISPs may face interoperability challenges between SDN-enabled and legacy devices. The fundamental conflict stems from protocol disparity [48]: SDN relies on OpenFlow programmability while legacy switches operate via BGP or OSPF. One potential solution is to employ middleware or adapters to bridge such protocol differences. For instance, Telekinesis [49] implemented a middleware to coordinate SDN switches in propagating seed packets containing preconfigured source MAC addresses toward legacy switches, thereby facilitating the update of their IP-MAC mappings. As illustrated in Fig. 5, this middleware architecture enables bidirectional cross-layer coordination by mediating communication between application-layer SFC deployment algorithms and the data plane in hybrid network infrastructures.

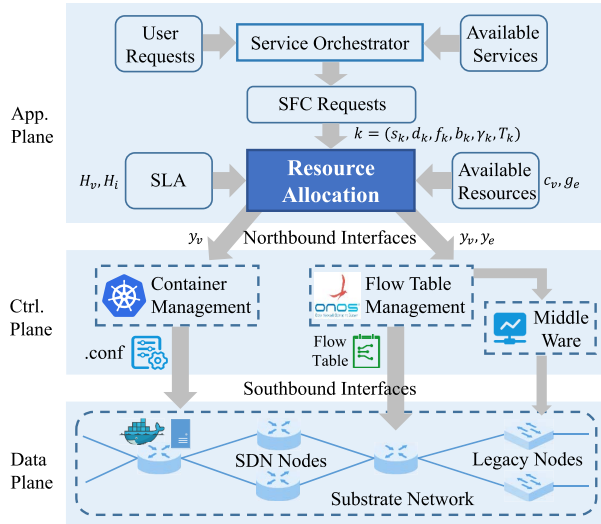

Fig. 5. Illustration of the practical implementation of RA.

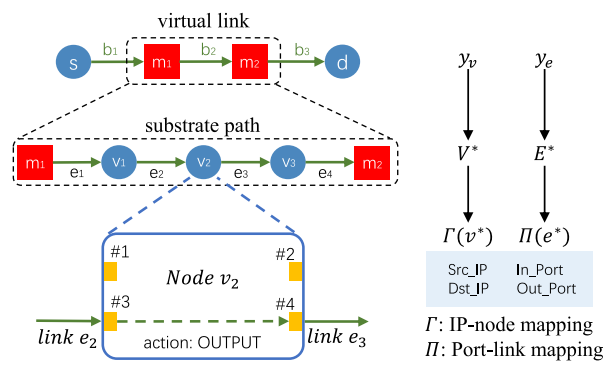

Fig. 6. Generating flow tables according to y_v and y_e.

V. Resource Provisioning

In this section, we formulate the RP problem, and an approach for solving RP is designed based on BO. Furthermore, an algorithm is proposed for the smooth switching between RP schemes.

A. Resource Provisioning Problem

When formulating the RP problem, the optimization objective can be chosen based on the ISP's practical requirements, such as maximizing benefit, minimizing delay, or maximizing QoS. In this article, the objective of RP is set to maximize *AccNum*, the number of accepted requests

$$AccNum = \sum_{k \in K_r} a_k^* \tag{50}$$

where a_k^* is the optimal solution obtained by solving SPR (36) given the resource capacity on each node and link

$$a_k^* = \mathrm{SPR}(K_r, c, g) \qquad \forall k \in K_r. \tag{51}$$

In addition, there is a constraint on the total budget for leasing network resources from the InP

$$\sum_{v \in V_s} \alpha_v \cdot c_v + \sum_{e \in E_s} \beta_e \cdot g_e \le C_0 \tag{52}$$

where C_0 denotes the ISP's total budget for resource leasing. Besides, limited by the capacity of physical networks, there are upper and lower limits on the leased resource on each node v and link e

$$c_v^{\min} \le c_v \le c_v^{\max} \qquad \forall v \in V_s \tag{53}$$

$$g_e^{\min} \le g_e \le g_e^{\max} \qquad \forall e \in E_s. \tag{54}$$

Summing up the preceding discussions, the RP problem is formulated as follows:

$$\begin{aligned} \max_{c,g} \quad & AccNum = \sum_{k \in K_r} a_k^* \\ \text{s.t.} \quad & \text{constraints} \quad (51)-(54) \\ \text{vars:} \quad & c \in R^V, \quad g \in R^E. \end{aligned} \tag{55}$$

Analogous to RA (46), RP (55) is characterized by severe nonlinearity. In constraint (51), the abstract function SPR is used to represent the derivation of a_k^* by solving the SPR problem, which leads to a lack of analytical expressions that directly relate a_k^* to (c, g). Despite the similarities above, there are several key distinctions between RP and RA. First, there is a significant gap in the dimension of optimization variables. In RA (46), $P \in R^{I \times E}$, while in RP (55), $(c, g) \in R^{V+E}$, which represents a reduction in the search space by an order of magnitude. Second, as outlined in Section I, the ISP solves RA for real-time and online deployment of SFC requests, whereas RP focuses on planning long-term resource leasing

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schemes. With a lower invocation frequency ranging from days to months, the demand for the execution speed of RP is correspondingly reduced. Generally, an execution time on the order of minutes or hours is sufficient to meet the latency requirements of RP.

In summary, compared to RA, RP exhibits a smaller search space and less stringent real-time demands, making it suitable for the adoption of DFO methods.

B. DFO Approach

BO is a state-of-the-art DFO technique that iteratively updates a surrogate model of the objective function and utilizes it as a guide to search for promising solutions. In this article, we propose resource provisioning based on Bayesian optimization (RPBO), a generalized framework that employs BO to solve RP (55). As shown in Algorithm 2, RPBO is elaborated as follows.

First, the search space $S \in R^{E+V}$ is defined (line 1), representing the capacity constraints on each node and link, i.e., (53) and (54) in the RP formulation. And *cons* represents other constraints that ISPs need to take into consideration, e.g., the constraint on the total cost (52). As depicted in line 3, the set of evaluation points, which is the network performance *Per* under each RP scheme x, is initialized based on the deployment history. Then, RPBO iteratively searches for better RP schemes. In each iteration, RPBO constructs the surrogate model $g(x)$ based on existing evaluation point set $D^{(n-1)}$ to approximate the complex function between x and *Per* (line 5), and updates the acquisition function $\alpha(x)$ (line 6). Subsequently, the RP scheme for the current iteration $x^{(n)}$ is determined by maximizing $\alpha(x)$, which characterizes the worthiness of exploring different RP schemes (line 7). $\alpha(x)$ automatically balances local searches in favorable regions of S with global searches in unexplored regions, achieving a harmonious integration of exploration and exploitation. By simulating the SFC deployment process, the performance $Per^{(n)}$ of the current RP scheme is evaluated (lines 8–12). Finally, as shown in line 13, $\{x^{(n)}, Per^{(n)}\}$ is added to the evaluation point set, which is then used to update $g(x)$ and $\alpha(x)$ in the next iteration.

During the evaluation of each RP scheme, it is necessary to simulate the SFC deployment to obtain the network performance $Per^{(n)}$ under a certain RP scheme $x^{(n)}$. One challenge here is that ISPs only have the request distribution RD, without having access to specific requests, as the planning of RP schemes occurs before the arrival of SFC requests. To address this issue, multiple experiments are conducted, where a set of requests K_r is randomly generated according to RD in each single experiment. Then we simulate the deployment of K_r to obtain an objective value $per^{(n,q)}$. Ultimately, the performance evaluation $Per^{(n)}$ of $x^{(n)}$ is determined by both the mean and variance of these objective values:

$$\mathrm{Per}^{(n)} = w_1 \cdot E\left(\mathrm{per}^{(n)}\right) + w_2 \cdot D\left(\mathrm{per}^{(n)}\right) \tag{56}$$

where w_1 and w_2 are adjustable weight parameters. Finally, as depicted in line 15, when the iteration ends, the optimal RP scheme x^* is selected from the iteration history.

Algorithm 2: Resource Provisioning Based on Bayesian Optimization (RPBO)

1 define search space $S \in R^{E+V}$, constraints *cons*;
2 initialize RP scheme: $x^{(0)} = \left[c^{(0)}, g^{(0)}\right] \in S$;
3 initialize the evaluation points set: $D^{(0)}$;
4 **for** $n = 1$ *to* N_{iter} **do**
5 construct surrogate model $g(x)$ according to $D^{(n-1)}$;
6 update the acquisition function $\alpha(x|D^{(n-1)})$;
7 obtain the RP scheme for current iteration: $x^{(n)} = argmax_x\{\alpha(x|D^{(n-1)})\}$;
8 **for** $q = 1$ *to* Q **do**
9 generate user requests K_r according to request distribution RD: $K_r = GenerateReq(RD)$;
10 simulatively deploy K_r by solving SPR (36): $per^{(n,q)} = \mathrm{SPR}(K_r, c^{(n)}, g^{(n)})$;
11 **end**
12 performance evaluation of the current RP scheme for request distribution RD: $Per^{(n)} = w_1 \cdot E(per^{(n)}) + w_2 \cdot D(per^{(n)})$;
13 augment the evaluation points set: $D^{(n)} = D^{(n-1)} \cup \{x^{(n)}, Per^{(n)}\}$;
14 **end**
15 obtain the optimal RP scheme from the iteration history: $x^* = x^{(n^*)}$, where $n^* = argmax_n\left\{Per^{(n)}\right\}$;

When utilizing BO to solve the RP problem, a crucial concern lies in the convergence speed. Evaluating the performance of an RP scheme is costly because multiple simulations for SFC deployment need to be conducted in each iteration of RPBO. Therefore, we hope to reduce the number of iterations required for RPBO to converge. In this article, instead of focusing on optimizations within BO itself, such as optimizing surrogate models and acquisition functions, we make more effective use of prior information to obtain a more refined search space (RSS) and select better initial solutions, thereby accelerating the convergence of RPBO. Intuitively, the ISP can evaluate the popularity of each link under a given RD, and narrow down the search spaces for unpopular links. Specifically, through multiple simulations of SFC deployment (lines 9 and 10 in Algorithm 2), the bandwidth usage on each link e is counted as the set $\{g_e^{\mathrm{use}}\}$. Due to the randomness rate in RD, $\{g_e^{\mathrm{use}}\}$ can be approximated to follow the Gaussian distribution $N(\mu_e, \sigma_e)$, where μ_e indicates the popularity of link e, and σ_e is negatively correlated with rate. Based on parameter estimation, the ISP can get μ_e and σ_e, then the initial capacity on each link e is set to μ_e, and the search space is set to $\mu_e \pm n\sigma_e$, where n is an adjustable parameter. In Section VI, we evaluate the effectiveness of this approach.

In summary, RPBO provides ISPs with a generalized framework that utilizes BO for capacity planning. The innovative adoption of BO allows ISP's effective decision-making to cope with service dynamics in SD-WAN environments.

C. Smooth Switching of RP Schemes

In this section, we focus on the strategies for ISPs to facilitate a smooth switch between RP schemes during the

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online deployment process. As shown in Algorithm 3, we address the following two issues.

First, a trigger mechanism is necessary to determine the specific conditions under which a switch for RP schemes should be initiated. Second, when there is a significant transition from the old request distribution RD_1 to the new one RD_2, it is possible that the new RP scheme $(c^{\text{new}}, g^{\text{new}})$ may become insufficient to support the existing active requests. Assume that at the switch moment, the bandwidth occupied on link e is g_e^{ocp}, and most of the load on link e originates from old requests that follow RD_1. In the new RP scheme, g_e^{new} calculated according to RD_2 may be smaller than g_e^{ocp}, making link e unable to carry the existing traffic load. Therefore, it is imperative to develop a method to promptly implement RP scheme switching while minimizing its impact on existing active requests.

For the first issue, it is necessary to measure the traffic pattern across the whole network. Here, we use a symmetric matrix $F \in R^{V\times V}$ to represent the flow distribution in network G. The load between node v_1 and v_2 is denoted as $F(v_1, v_2) = \sum_{k\in K_{12}} b_k$, where b_k denotes the data rate of request k, and K_{12} denotes the set of requests that originate from v_1 and terminate at v_2, or vice versa. In practical scenarios, the flow distribution F can be obtained through the operational support system (OSS). As shown in lines 4–6 of Algorithm 3, at each time interval $t = nT$, the ISP obtains current flow distribution F_n and calculates its secondary moment S_n. Then the variation in flow distribution is derived as

$$\Delta S_n = \left[\sum_{i=1}^{V}\sum_{j=1}^{V} \Delta F_n^2(i,j)\right]^{\frac{1}{2}} \tag{57}$$

where $\Delta F_n = F_n - F_{n-1}$. As depicted in lines 8 and 9, if $\Delta S_n/S_n$ is detected greater than threshold α, it indicates that a significant change has occurred in the flow distribution, making the existing RP scheme suboptimal for the current RD, and RPBO needs to be invoked to calculate the new RP scheme $(c^{\text{new}}, g^{\text{new}})$.

Regarding the second issue, backup resources are introduced to ensure the existing active requests during the switching process. As indicated in line 10, after calculating $(c^{\text{new}}, g^{\text{new}})$, the bandwidth deficiency Δg_e on each link e is counted as

$$\Delta g_e = \begin{cases} g_e^{ocp} - g_e^{\text{new}}, & \text{if} \quad g_e^{\text{new}} < g_e^{ocp} \\ 0, & \text{if} \quad g_e^{\text{new}} \geq g_e^{ocp}. \end{cases} \tag{58}$$

Based on Δg_e, the substrate link set E_s is partitioned into two subsets: E^+ denotes the links where $\Delta g_e = 0$, and E^- denotes the links where $\Delta g_e > 0$. For links $e \in E^+$, as shown in line 12, the new RP scheme can be directly applied. While for links $e \in E^-$, backup resources need to be introduced. Assuming that the ISP can prepare a maximum of β fraction of the total network resources as backup $(0 < \beta < 1)$. If the total bandwidth deficiency $\sum_{e\in E_s} \Delta g_e$ is below this threshold, the capacity of each link $e \in E^-$ is adjusted upward from g_e^{new} to g_e^{ocp} to compensate for the bandwidth deficiency and ensure all active requests carried on it (lines 14–16). Note that the backup resources allocated to each link $e \in E^-$ are exclusively used to

Algorithm 3: Smooth Switch of RP Schemes

1 initialization: set timer $t = 0$, set interval T, get flow distribution $F_0 \in R^{V\times V}$;
2 main loop:
3 **while** $t = nT$ **do**
4 get current flow distribution F_n;
5 calculate $S_n = \left[\sum_{i=1}^{V}\sum_{j=1}^{V} F_n^2(i,j)\right]^{\frac{1}{2}}$;
6 calculate ΔS_n, the variation in flow distribution;
7 release the idle backup resource on each $e \in E^-$;
8 **if** $\Delta S_n/S_n \geq \alpha$ **then**
9 invoke RPBO, update the RP scheme, get $(c^{\text{new}}, g^{\text{new}})$;
10 get Δg_e on each link e;
11 **for** *each link* $e \in E^+$ **do**
12 apply new RP scheme: $g_e \leftarrow g_e^{\text{new}}$;
13 **end**
14 **if** $\sum_{e\in E_s} \Delta g_e \leq \beta \cdot \sum_{e\in E_s} g_e^{\text{new}}$ **then**
15 **for** *each link* $e \in E^-$ **do**
16 Employ backup resources to ensure existing requests: $g_e \leftarrow g_e^{ocp}$;
17 **end**
18 **end**
19 **else**
20 **for** *each link* $e \in E^-$ **do**
21 get weight u_e for each link e;
22 proportionally allocate the backup resource: $g_e \leftarrow g_e + u_e\beta \cdot \sum_{e\in E_s} g_e^{\text{new}}$;
23 update $\Delta g_e = g_e^{ocp} - g_e$;
24 **end**
25 solve the DR problem (61), terminate several active SFCs;
26 **end**
27 **end**
28 **end**

ensure existing requests and cannot be utilized to deploy newly arrived requests. As indicated in line 7, after the termination of each existing request, the corresponding backup resources should be released.

Another instance arises when it is not possible to ensure all existing active requests even with all backup resources invested, i.e., $\sum_{e\in E_s} \Delta g_e > \beta \cdot \sum_{e\in E_s} g_e^{\text{new}}$. In this case, some active SFCs have to be terminated. Specifically, as shown in line 21, the weight u_e is calculated, representing the ratio of the bandwidth deficiency on link e to the total deficiency

$$u_e = \frac{\Delta g_e}{\sum_{e\in E_s} \Delta g_e} \quad \forall e \in E^-. \tag{59}$$

Then, all backup resources are allocated proportionally to each link $e \in E^-$ based on the weight u_e, and the bandwidth deficiency Δg_e is then updated (lines 22 and 23).

Finally, as shown in line 25, the ISP needs to determine which active requests to terminate by solving the DownRequest (DR) problem. Assuming there are a total of Q active requests, the boolean variable $d_q = 1$ if the qth request

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should be terminated. $G_{\text{use}} \in R^{E\times Q}$ and $G_{\text{use}}(e, q)$ denotes the load generated by the qth request on link e. To eliminate the bandwidth deficiency, the amount of released bandwidth on each link $e \in E^-$ should be no less than its deficiency, as indicated by the constraint below

$$\sum_{q=1}^{Q} G_{\text{use}}(e, q) \cdot d_q \geq \Delta g_e \quad \forall e \in E^-. \tag{60}$$

To minimize the impact of RP scheme switching on existing requests, the optimization objective of DR is set to minimize the weighted sum of terminated requests, i.e., $\sum_{q=1}^{Q} d_q \cdot w_q$, where the weight w_q represents the importance or priority of the qth active request. In summary, the DR problem is formulated as follows:

$$\begin{aligned} \min_{d} \quad & N_{\text{down}} = \sum_{q=1}^{Q} d_q \cdot w_q \\ \text{s.t.} \quad & \text{constraints} \quad (58),\ (60) \\ \text{vars:} \quad & d_q \in \{0, 1\} \quad \forall q = [1, Q]. \end{aligned} \tag{61}$$

By solving DR (61), the ISP ensures that the termination of active requests is done in a way that maintains the integrity of the most vital services, thus minimizing the overall disruption of service quality.

Although Section I demonstrates that service request distributions in private WANs generally exhibit gradual evolutionary patterns, practical deployments may still encounter extreme scenarios characterized by abrupt and/or drastic RD shifts. Here, we analyze the adaptability mechanisms for handling such exceptional conditions and reveal two inherent tradeoffs. First, to address highly dynamic traffic variations, Algorithm 3 requires parameter adaptation (e.g., reducing α in line 8) to enhance monitoring sensitivity for RD and minimize RP scheme switching delay. This adjustment introduces a fundamental tradeoff between switching responsiveness and backup resource utilization: earlier switching necessitates greater backup resources to maintain more existing active requests from legacy RD_1. Second, drastic changes in RD may lead to insufficient backup resources, causing forced termination of some active requests. One potential solution is soft handover, which gradually adjusts the RP scheme to $(c^{\text{new}}, g^{\text{new}})$ over several time slots. While soft switching helps maintain existing active requests, the slower network adjustment may affect the acceptance rate of incoming requests (mostly from RD_2). Therefore, given limited backup resources, a tradeoff is necessary between preserving the existing active requests and the acceptance of new-coming ones. In Section VI-C, simulations of RP scheme switching are conducted to illustrate the above tradeoffs.

VI. Simulation Results

In this section, simulation results are presented to evaluate the proposed approaches. For RA, the performance of IBD is evaluated by comparing it with three baselines. For RP, we evaluate the convergence of RPBO under different constraints and verify its effectiveness by simulating an alteration in RD.

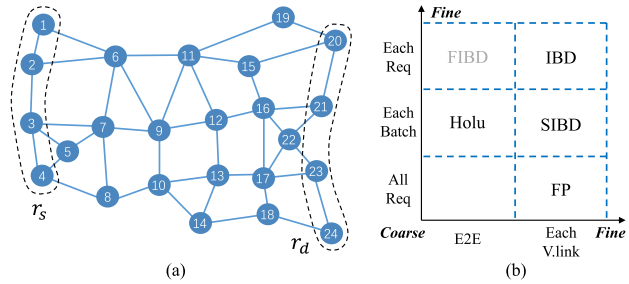

Fig. 7. Illustration of (a) the 24-node US mesh network, and (b) routing granularity comparison.

A. Experimental Setup

Network Parameters: As shown in Fig. 7(a), the 24-node US mesh network [50] is adopted as $G(V_s, E_s)$. By default, α_v and β_e are set to 1 for each node and link, c_v is set to 3000 for each $v \in V_s$, and g_e is randomly generated within [1500, 3500], representing a network with abundant computing resource and constrained bandwidth. The delay experienced on each node $\tau_v = 0.1$, and the link delay τ_e is randomly generated within [1, 10] for each $e \in E_s$. The significant disparity between τ_v and τ_e indicates that the link delay accounts for the dominant part of the E2E delay of an SFC request. To simulate the constraints imposed on SFC deployment by hardware limitations and SLAs, each column of H_v and H_i is randomly generated, with 80% of the elements set to 1 and 20% set to 0. This indicates that for each kind of service $m \in M_s$, 20% of the substrate nodes are not eligible for its deployment.

Service Parameters: According to the request distribution RD, a set of SFC requests are generated with the same chain structure but different source s_k and destination d_k. There are four sequentially connected services in each SFC, and the amount of computing resource consumed by each service $\xi_k(l) = 1, l = [1, 4]$. The data rate of each virtual link $b_k(l) = 100, l = [1, 4]$. The revenue of deploying request k is $\gamma_k = 400 + 100 * \text{Dist}(s_k, d_k)$, where $\text{Dist}(s_k, d_k)$ is the hop count of the shortest path between s_k and d_k. The maximum allowable E2E delay T_k is set to 25 for each request k. For each individual simulation, 200 requests arrive sequentially and get deployed in batches. By default, the request batch size K is set to 10.

B. Evaluation for RA

1) Benchmark Approaches: As previously discussed, IBD solves RA with inner and outer iterations of the routing matrix P. To evaluate the effectiveness of these iterations, the following approaches are compared.

1) *IBD:* Our approach for solving RA, where P is updated by both inner and outer iteration. The weights w_1 and w_2 in Algorithm 1 are set to 0.8 and 0.2, respectively, and $n_{\max}$ is set to 3.
2) *SIBD:* Simplified from IBD by removing the inner iterations. As elaborated in Section VI-C, SIBD captures the worst-case performance of IBD under degenerated routing conditions, thus serving as a performance-bound metric of IBD.

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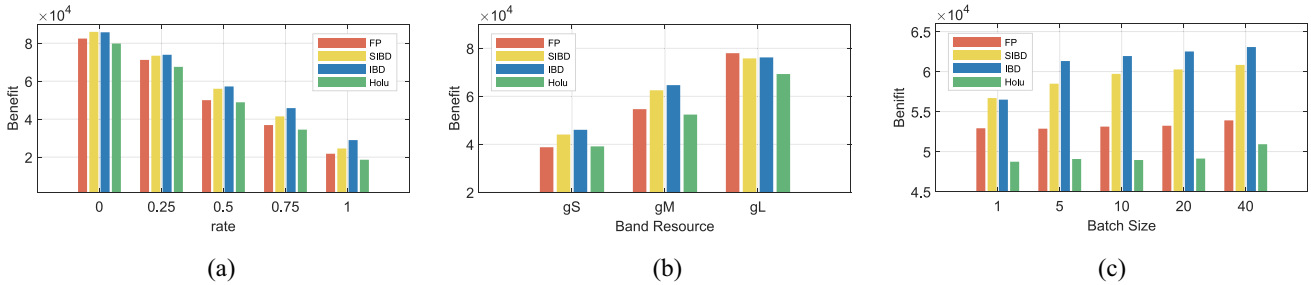

Fig. 8. Comparison of the ISP's benefit across different (a) request distributions, (b) bandwidth capacities, (c) deployment granularities.

TABLE II
RANGES OF LINK CAPACITY FOR THREE TEST NETWORKS

Network	Range of g_e
gS	$500 \sim 2500$
gM	$1500 \sim 3500$
gL	$3000 \sim 5000$

3) *FixedPath (FP):* Representing the simplification of *routing* in [20], [21], this approach has neither inner nor outer iteration of the routing matrix, that is, P remains unchanged during the whole deployment process.
4) *Holu:* This approach is derived from the algorithm proposed in [18], where *placement* and *routing* are successively solved using heuristic methods.

Fig. 7(b) compares the routing granularity of the above approaches. Horizontally, for IBD and its simplified versions SIBD and FP, the routing matrix specifies the optional paths between adjacent services in each SFC, that is, *routing* is implemented at the granularity of each virtual link within the SFC. While Holu operates with a coarser routing granularity, as it calculates E2E paths for routing each batch of SFC requests. Vertically, IBD (with inner and outer iterations), SIBD (with only outer iterations), and FP (P remains unchanged) can be regarded as routing for each individual request, each batch of requests, and all requests, respectively.

2) Numerical Results: First, as the optimization objective of SPR (36), the ISP's benefit is evaluated using the approaches described above. As shown in Fig. 8, we evaluate the ISP's benefit from the following three aspects.

1) *Request Distribution:* Fig. 8(a) shows the ISP's benefit under different rate of RD. rate characterizes the regularity of a request batch. Given the source and destination region r_s, r_d, the decrease in rate implies an irregular traffic pattern of K_r.
2) *Bandwidth Resource:* Fig. 8(b) shows the ISP's benefit in networks with different bandwidth capacities. As shown in Table II, the link capacity g_e is randomly generated within three ranges. In gS and gM, the network is bandwidth-constrained, while gL has abundant bandwidth to deploy all SFC requests.
3) *Deployment Granularity:* Fig. 8(c) shows the ISP's benefit with the batch size K changing from [1, 5, 10, 20, 40]. K indicates the deployment granularity of SFC requests.

Based on Fig. 8(a), two conclusions can be drawn. Vertically, with the increase of rate from 0 to 1, the ISP's benefit experienced a reduction of 70%. This is because a more concentrated RD aggravates network congestion, thus decreasing the acceptance rate and ISP's benefit. Horizontally, given a certain rate, IBD achieved higher benefits compared to the three baselines, and this gap widens as rate increases. This indicates that for a batch of requests with concentrated distribution, IBD can alleviate network congestion and improve the ISP's benefit. In Fig. 8(b), it is evident that the ISP's benefit is positively correlated with bandwidth capacity. In gS and gM, IBD achieved the highest benefit. While in gL, FP achieved a slightly higher benefit than IBD. The reason for this anomaly is explained as follows. In the case of abundant bandwidth (gL), FP has the same acceptance rate as IBD (near 100%) but lower bandwidth overhead, as FP always adopts the minimum hop path for routing between services, making it occupy less bandwidth. In Fig. 8(c), the increase of K leads to higher benefits for the ISP. With the outer iteration, SIBD and IBD perform better than FP. Besides, with the inner iteration balancing the data flow from different requests, IBD performs better than SIBD as long as $K > 1$.

In addition, the performance gap between IBD and its theoretical lower-bound benchmark SIBD exhibits a widening disparity under the following conditions: 1) higher rate; 2) more constrained bandwidth capacities; and 3) higher K. These findings collectively validate the efficiency of IBD's load-balancing mechanisms in complex network environments. While Holu underperforms IBD, SIBD, and FP due to coarser routing granularity, as its fixed E2E path computation reduces complexity at the expense of flexibility in bandwidth-constrained networks.

Second, we compare the acceptance rates during the deployment process, with K set to 20. To analyze the impact of dynamic network conditions on algorithm performance, network failures were introduced. Specifically, before the deployment of each batch of requests, each link $e \in E_s$ has a 5% probability of failure, and failed links will be unable to carry the incoming SFC data flows. Given the batch number n, we can calculate $\mathrm{AR}(n)$, the acceptance rate of batch n, and $\mathrm{CAR}(n)$, the cumulative acceptance rate up to the nth batch

$$\mathrm{AR}(n) = \frac{\mathrm{AccNum}(n)}{K} \quad , \quad \mathrm{CAR}(n) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{AR}(i)$$

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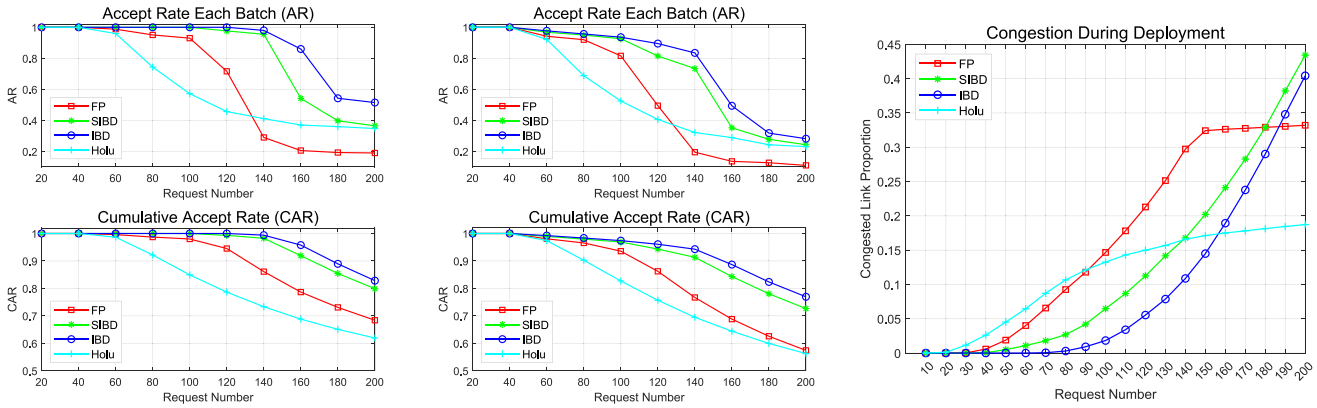

(a) (b) (c)

Fig. 9. Comparison of three indicators during the deployment process. (a) Acceptance rate AR and CAR without network failure. (b) AR and CAR considering network failure. (c) congested link rate.

where $AccNum(n)$ denotes the number of accepted requests in batch n. Fig. 9(a) and (b) compares AR and CAR under conditions with and without network failures. In the early deployment stages, FP's AR was slightly lower than that of IBD and SIBD. However, in the later stages, FP's AR dropped significantly. Due to its coarse routing granularity, Holu exhibited the lowest AR and CAR. In terms of rejection rate, IBD achieved a 50% reduction compared to Holu. After introducing network failures, the acceptance rates of all four approaches decreased, with FP showing the largest decline (from 68.4% to 57.3%). This is because the routing matrix remains unchanged during the deployment process, preventing FP from adapting to dynamic changes of network environment.

Third, the performance of load balancing is evaluated by comparing the congested link rate (CLR), which indicates the congestion degree across the whole network. Specifically, *CLR* depends on the bandwidth utilization θ_e of each link e

$$\mathrm{CLR} = \frac{\sum_{e \in E_s} f(\theta_e)}{|E_s|}$$

where f is an activation function defined as follows:

$$f(\theta) = \begin{cases} 0, & \text{if} \quad 0 \le \theta \le 0.5 \\ 2\theta - 1, & \text{if} \quad 0.5 < \theta \le 1. \end{cases}$$

As depicted in Fig. 9(c), during the early deployment stage (request No.10 $\sim$ 150), IBD produced the lowest congestion by adopting both the inner and outer iterations and accepted more requests in the later stage (request No.150 $\sim$ 200). At this stage, FP and Holu stopped growing in *CLR* because they failed to accept more incoming requests due to the severe congestion on popular links. Therefore, the effectiveness of IBD in alleviating link congestion can be verified.

Finally, the complexity and scalability of the proposed algorithms are analyzed by comparing the execution time. As shown in Fig. 10, the execution time required for IBD, FP, and Holu to deploy a single batch of requests are compared under different SFC lengths L, batch sizes K, and network sizes V. Here, we use the WS small-world model [51] to generate random network topologies of different scales. On one hand, as the complexity of SPR is $O(K \cdot L \cdot V^2)$, the

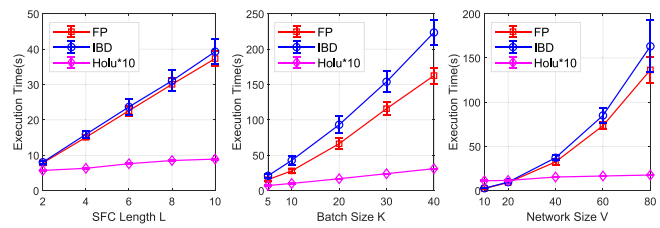

Fig. 10. Execution time with different (a) SFC length L, (b) batch size K, (c) network size V.

execution time of FP and IBD is observed proportional to K, L, and V^2 in Fig. 10, with IBD exhibiting more fluctuation. This is due to the probabilistic load balancing in IBD, which causes uncertainty in the number of internal iterations. On the other hand, the coarse routing granularity of Holu results in a significantly lower complexity compared to FP and IBD. The execution time of Holu is nearly independent of the network scale, giving it stronger scalability for deploying SFCs in large-scale networks.

C. Evaluation for RP

In the simulations for RPBO, considering the bandwidth-constrained characteristic of SD-WAN, all nodes in G are assigned with ample computing resources, and the optimization variables are reduced to the bandwidth capacities $g \in R^E$.

1) Convergence Evaluation of RPBO: In this article, RPBO is implemented using OpenBox [52], an open-source, general-purpose black-box optimization framework. To accelerate the convergence of RPBO, the following two methods are adopted and compared.

RSS: As mentioned in Section V-B, we accelerate the convergence of RPBO by utilizing a more RSS derived from the bandwidth usage distribution on each link. Tables III and IV show the original search space (OSS) and the RSS.

Transfer Learning (TL): OpenBox supports the use of TL for convergence acceleration. In this article, an initial surrogate model is trained using ten sets of historical iteration data. For

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TABLE III
OSS

Link Number	Initial bandwidth	Searching Range
1 4 6 27 30	5600	4000 ∼ 8000
7 13 24	3000	2000 ∼ 4000
else	1500	500 ∼ 2500

TABLE IV
RSS

Link Number	Initial bandwidth	Searching Range
6 27	6700	5000 ∼ 7500
1 4 30	5000	4000 ∼ 6000
7 13 24	3350	2000 ∼ 4000
14 15 17 20 25	1650	1500 ∼ 3000
else	1000	500 ∼ 1500

the sake of controlling variables, the experiments in TL adopt the OSS.

With the budget C_0 in (52) set to 72000, 200 SFC requests were serially deployed using FP and IBD. Focusing on the request rejection number, the convergence of RSS and TL are compared with the baseline using OSS. Each experiment is repeated 30 times, the mean and variance of the iteration histories are visualized in Fig. 11. First, comparing RSS/TL with OSS, it is observed that RPBO achieved faster convergence and less fluctuation in RSS/TL with the guidance of extra prior information. As for the performance gap between RSS/TL and OSS, IBD has a smaller gap than FP, mainly because IBD can achieve a lower request rejection rate under the same RP scheme. Second, comparing RSS with TL, it is evident that RSS outperforms TL when using the FP algorithm, but the performance of the two is almost identical when using the IBD algorithm. This is attributed to the fact that IBD possesses stronger nonlinearity and uncertainty. Therefore, to accelerate the convergence of RPBO, RSS is more suitable for the more linearly inclined algorithms like FP, while TL offers a stronger universality.

Besides, the convergence of RPBO using FP and IBD is analyzed under different budget constraints, where C_0 ranges from 65000, 72000, and 80000. As shown in Fig. 12, with the increase of C_0, the convergence of RPBO experienced an increase in speed along with a decrease in fluctuation. Meanwhile, the performance gap between FP and IBD widens as C_0 decreases. This indicates that a bidirectional adaptation between the network and services is achieved when applying IBD in RPBO, enabling RPBO to achieve better convergence performance, especially in bandwidth-constrained networks.

2) Simulations of RP Scheme Switching: First, to verify the impact of request distribution RD on network performance, three types of RD were simulated, with different r_s, r_d, and the same rate. Focusing on the request acceptance rate, 200 SFC requests were deployed under different RD with the same RP scheme. As shown in Fig. 13, vertically, for a given RP scheme, there is a significant variation in network performance under different r_s and r_d. Horizontally, the fluctuation of the acceptance rate for IBD (68% to 81%) is smaller than FP (54% to 77%) and Holu (51% to 72%). This indicates that IBD can partially reduce network performance fluctuations caused by

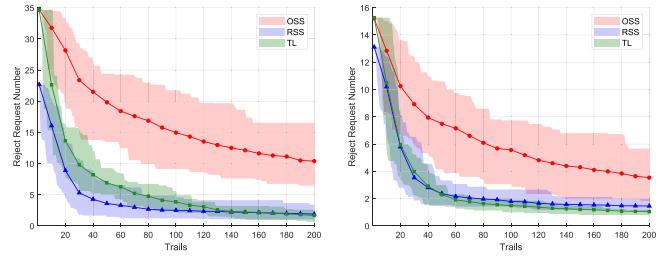

Fig. 11. Comparison of convergence performance using (a) FixedPath, (b) IBD.

service dynamics, as the adaptation of services to the network is implemented in IBD.

Second, the effectiveness of Algorithm 3 is validated by simulating an arrival-departure process of SFC requests, where the arrival rate $K = 10$ and the departure rate is 9. The number of active SFCs (N_a) is used to measure the network performance. When Req.id = 800, the request distribution shifts from RD_1 to RD_2. To verify the relationship between the responsiveness of RP scheme switching and its backup resource overhead, different switching delays d were simulated, with $d = 200$ in Fig. 14 and $d = 0$ in Fig. 15. From the red dotted line in Fig. 14, it is observed that N_a experienced a decline of 20% if no switch in the RP scheme was made. After implementing the switching using Algorithm 3, N_a gradually recovered to the level before the RD change. The comparison of Figs. 14 and 15 reveals that to achieve a smooth switching with zero forced request termination, the required backup resource amount β^* increases as d decreases, as $0.05 \leq \beta^* \leq 0.15$ when $d = 200$ and $0.25 \leq \beta^* \leq 0.35$ when $d = 0$. This validates the fundamental tradeoffs analyzed in Section V-C. In conclusion, the switch of RP schemes enables the network to adapt to services, thus reducing the network performance fluctuations caused by service dynamics.

VII. Conclusion and Future Work

In SD-WANs, due to the bandwidth-constrained characteristics of the network and the dynamics of services, it is crucial to address both RA and RP, which complement each other, representing the bidirectional adaptation between the network and services. In this work, RA and RP are unified within one framework. For RA, we proposed the IBD algorithm, where load balancing of data flows is achieved in a probabilistic manner. For RP, we innovatively adopted BO to design RPBO, a generalized framework to assist ISPs in planning the resource leasing schemes. To our knowledge, this is the first work to apply BO to RP problems. Simulation results indicate that IBD can reduce link congestion during the SFC deployment process, and improve the request acceptance rate and the ISP's benefit. Besides, the effectiveness of RPBO has been validated, demonstrating its ability to converge within an acceptable time frame, and reduce the performance fluctuations resulting from service dynamics.

Our future work will focus on reducing the computational complexity of IBD. As mentioned in Section IV-C, we aim to design a faster version of IBD (FIBD), whose routing granularity is shown in Fig. 7(b), and further optimization on

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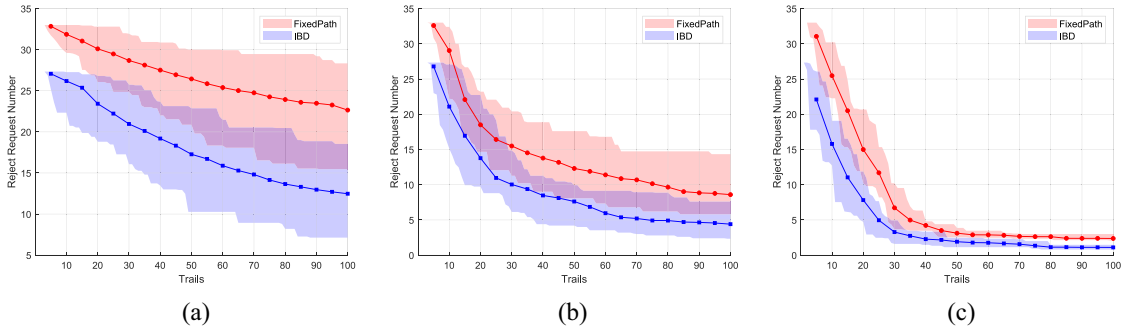

Fig. 12. Comparison of convergence performance under different budget constraints. (a) $C_0 = 65000$. (b) $C_0 = 72000$. (c) $C_0 = 80000$.

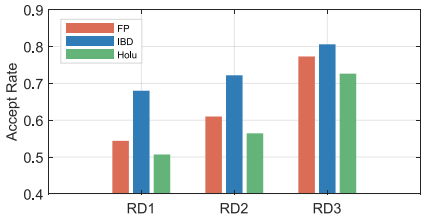

Fig. 13. Acceptance rates under different request distributions.

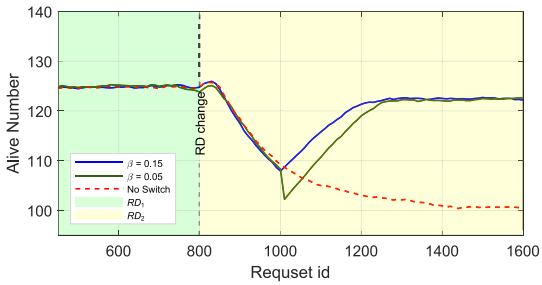

Fig. 14. Alive SFC number before and after the request distribution change, with switching delay $d = 200$.

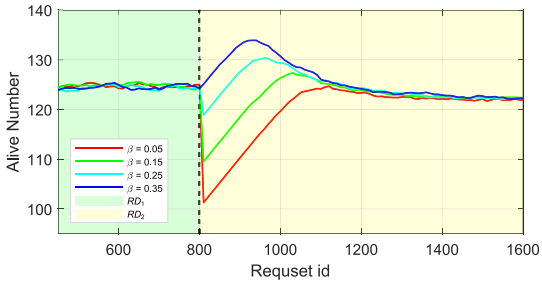

Fig. 15. Alive SFC number before and after the request distribution change, with switching delay $d = 0$.

the load balancing mechanism is needed. Besides, we will develop a soft-switching method to address drastic RD shifts, and further minimize the impact of RP scheme switching on overall service quality.

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